Aptitude & Reasoning Study Guide: Solutions

A. Percentages & Profit/Loss

Q1. A number is increased by 25% and then decreased by 20%. Find the net percentage change.

Solution: Let the original number be 100. After a 25% increase, it becomes 100 × 1.25 = 125. After a 20% decrease, it becomes 125 × (1 - 0.20) = 125 × 0.8 = 100. The net percentage change is **0%**.

Q2. A student scores 30% below the pass mark. After gaining 40 additional marks, the student scores 10% above the pass mark. If the pass mark is 40% of the maximum marks, find the maximum marks.

Solution: Let the pass mark be P. The difference between 10% above P and 30% below P is 10% - (-30%) = 40% of P. We are given this difference is 40 marks, so 40% of P = 40 $\Rightarrow$ P = 100. Since the pass mark is 40% of the maximum marks (M), 0.40 × M = 100 $\Rightarrow$ M = 100 / 0.4 = **250**.

Q3. An article is sold at a 15% loss. If it were sold for ■255 more, the profit would be 10%. Find the cost price.

Solution: The change from a 15% loss to a 10% profit is a net change of 15% + 10% = 25% of the Cost Price (CP). Given that 25% of CP = ■255, we find CP = ■255 × 4 = **■1,020**.

Q4. A dishonest dealer uses a 900 g weight instead of 1 kg and charges the cost price of 1 kg. Find the profit percentage.

Solution: The dealer gives 900 g but charges for 1000 g. Profit = 1000g - 900g = 100g. Profit percentage is calculated on the actual weight given: (100 / 900) × 100% = 1/9 × 100% = **11.11%**.

B. Ratio, Average & Ages

Q5. The incomes of A and B are in the ratio 5:7, and their expenditures are in the ratio 2:3. If both save ■4,000, find their incomes.

Solution: Let their incomes be 5x and 7x, and expenditures be 2y and 3y. Savings = Income - Expenditure $\Rightarrow$ 5x - 2y = 4000 and 7x - 3y = 4000. Solving these simultaneous equations: multiply the first by 3 and the second by 2: 15x - 6y = 12000 and 14x - 6y = 8000. Subtracting gives x = 4000. Thus, A's income = 5 × 4000 = **■20,000** and B's income = 7 × 4000 = **■28,000**.

Q6. The average age of 30 students is 20 years. When a teacher joins, the average becomes 21 years. Find the teacher's age.

Solution: Total age of 30 students = 30 × 20 = 600 years. Total age of 31 people (students + teacher) = 31 × 21 = 651 years. Teacher's age = 651 - 600 = **51 years**.

Q7. The present ages of a father and son total 60 years. Six years ago, the father was five times as old as the son. Find their present ages.

Solution: Let the present ages of the father and son be F and S. We know $F + S = 60$. Six years ago, $(F - 6) = 5(S - 6) \Rightarrow F - 6 = 5S - 30 \Rightarrow F - 5S = -24$. Substituting $F = 60 - S$ yields: $60 - S - 5S = -24 \Rightarrow 84 = 6S \Rightarrow S = 14$. Therefore, the son's present age is **14 years** and the father's present age is 60 - 14 = **46 years**.

C. Time & Work

Q8. A can complete a job in 12 days and B in 18 days. They work together for three days, after which C joins. C alone can complete the job in 24 days. How many additional days are needed to finish the remaining work?

Solution: Let total work be the LCM of 12, 18, 24 = 72 units. Efficiencies: A = 72/12 = 6 units/day, B = 72/18 = 4 units/day, C = 72/24 = 3 units/day. In 3 days, A and B complete: 3 × (6 + 4) = 30 units. Remaining work = 72 - 30 = 42 units. Now A, B, and C work together with combined efficiency = 6 + 4 + 3 = 13 units/day. Additional days needed = 42 / 13 = **3 3/13 days** (or approx. 3.23 days).

Q9. Twelve workers can complete a project in 20 days. After five days, four workers leave. How many additional days will the remaining workers need?

Solution: Total work = 12 × 20 = 240 worker-days. In 5 days, work done = 12 × 5 = 60 worker-days. Remaining work = 240 - 60 = 180 worker-days. Remaining workers = 12 - 4 = 8 workers. Additional days required = 180 / 8 = **22.5 days**.

D. Time, Speed & Distance

Q10. A train 180 m long crosses a pole in 9 seconds and a platform in 21 seconds. Find the platform's length.

Solution: Speed of the train = Length of train / time to cross pole = 180 m / 9 s = 20 m/s. To cross a platform of length P, the train covers a total distance of $(180 + P)$ in 21 seconds. So, $180 + P = 20 \times 21 \Rightarrow 180 + P = 420 \Rightarrow P =$ **240 m**.

Q11. A person walks to work at 5 km/h and arrives 12 minutes late. At 6 km/h, the person arrives 10 minutes early. Find the distance.

Solution: Let the distance be d km. The time difference between being 12 mins late and 10 mins early is 22 minutes = 22/60 hours. Difference in time: $(d/5) - (d/6) = 22/60 \Rightarrow d/30 = 22/60 \Rightarrow d = 30 \times 22/60 =$ **11 km**.

Q12. Two trains of lengths 150 m and 200 m travel in opposite directions at 54 km/h and 72 km/h. Find the time taken to cross each other.

Solution: Total distance to cover = 150 + 200 = 350 m. Relative speed in opposite directions = 54 + 72 = 126 km/h. Convert to m/s: 126 × 5/18 = 35 m/s. Time taken = 350 m / 35 m/s = **10 seconds**.

E. Number System & Algebra

Q13. Find the remainder when $7^{1}\blacksquare^{3}$ is divided by 13.

Solution: Using Fermat's Little Theorem, $7^{12} \equiv 1 \pmod{13}$. We can write 103 as 12 × 8 + 7. Therefore, $7^{103} = (7^{12})^{8} \times 7^{7} \equiv 1^{8} \times 7^{7} \equiv 7^{7} \pmod{13}$. Let's calculate powers of 7 mod 13: $7^{1} \equiv 7$, $7^{2} = 49 \equiv 10$, $7^{3} \equiv 70 \equiv 5$, $7^{4} \equiv$

$35 \equiv 9$, $7^6 = (7^3)^2 = 25 \equiv 12 \equiv -1$. Thus, $7^7 = 7^6 \times 7^1 \equiv -1 \times 7 = -7 \equiv$ **6**. The remainder is 6.

Q14. Find the number of trailing zeros in 125!.

Solution: Trailing zeros are determined by the number of factors of 5 in 125!. Using Legendre's formula: ■125/5■ + ■125/25■ + ■125/125■ = 25 + 5 + 1 = **31 trailing zeros**.

Q15. If $x + 1/x = 5$, find $x^2 + 1/x^2$ without solving for x.

Solution: Square both sides of the equation: $(x + 1/x)^2 = 5^2 \Rightarrow x^2 + 2(x)(1/x) + 1/x^2 = 25 \Rightarrow x^2 + 2 + 1/x^2 = 25 \Rightarrow x^2 + 1/x^2 = 25 - 2 =$ **23**.

Q16. If $a + b = 12$ and $ab = 32$, find $a^3 + b^3$.

Solution: Using the algebraic identity: $a^3 + b^3 = (a + b)(a^2 - ab + b^2) = (a + b)[(a + b)^2 - 3ab]$. Substituting the given values: $a^3 + b^3 = 12 \times [12^2 - 3(32)] = 12 \times [144 - 96] = 12 \times 48 =$ **576**.

F. Probability & Combinations

Q17. Two fair dice are rolled. Find the probability that their sum is 8.

Solution: Total outcomes when rolling two dice = 6 × 6 = 36. Favourable pairs for a sum of 8 are: (2,6), (3,5), (4,4), (5,3), and (6,2) — a total of 5 outcomes. Probability = **5/36**.

Q18. A bag contains 5 red, 4 blue, and 3 green balls. Two balls are drawn without replacement. Find the probability that both are the same colour.

Solution: Total balls = 5 + 4 + 3 = 12. Total ways to choose 2 balls = $^{12}C_2 = (12 \times 11) / 2 = 66$. Ways to draw 2 balls of the same color: Red = $^5C_2 = 10$; Blue = $^4C_2 = 6$; Green = $^3C_2 = 3$. Total favourable ways = 10 + 6 + 3 = 19. Probability = **19/66**.

Q19. How many distinct arrangements can be made using all the letters of MISSISSIPPI?

Solution: The word MISSISSIPPI has 11 letters in total, with repetitions: M=1, I=4, S=4, P=2. The number of distinct arrangements is given by 11! / (4! × 4! × 2!) = 39,916,800 / (24 × 24 × 2) = 39,916,800 / 1152 = **34,650**.

Q20. In how many ways can a committee of four be selected from six men and five women if it must contain at least two women?

Solution: We evaluate the valid scenarios:
1) 2 Women & 2 Men: $^5C_2 \times {}^6C_2 = 10 \times 15 = 150$
2) 3 Women & 1 Man: $^5C_3 \times {}^6C_1 = 10 \times 6 = 60$
3) 4 Women & 0 Men: $^5C_4 \times {}^6C_0 = 5 \times 1 = 5$.
Total combinations = 150 + 60 + 5 = **215 ways**.

G. Mixtures & Interest

Q21. A 50-litre solution contains 30% acid. How much pure acid must be added to make the solution 50% acid?

Solution: Initial acid volume = 30% of 50 = 15 litres. Volume of water remains constant at 50 - 15 = 35 litres. In the final 50% solution, water must comprise 50% of the total volume. Thus, final total volume = 35 / 0.5 = 70 litres. Pure acid to be added = 70 - 50 = **20 litres**.

Q22. A sum of ■20,000 is invested at 10% per annum for two years. Find the difference between compound interest and simple interest, assuming annual compounding.

Solution: For a 2-year period, the difference between CI and SI is given by the formula: Difference = $P(r/100)^2$. Plugging in values: Difference = 20000 × $(10/100)^2$ = 20000 × 0.01 = **■200**.

H. Logical Reasoning

Q23. Find the next number: 3, 8, 18, 38, 78, ?

Solution: The pattern is that each term is multiplied by 2 and then 2 is added: (3 × 2) + 2 = 8; (8 × 2) + 2 = 18; (18 × 2) + 2 = 38; (38 × 2) + 2 = 78. The next number is (78 × 2) + 2 = 156 + 2 = **158**.

Q24. Five people T, R, P, Q, and S sit in a row in that order. Who sits in the middle?

Solution: Given the seating layout is explicitly listed as T, R, P, Q, S in that order, the item right in the middle (third from either side) is **P**.

Q25. All developers are logical thinkers. Some logical thinkers are analysts. Can we conclude that some developers are analysts? Explain.

Solution: No, we cannot definitely conclude this. The group of 'developers' is fully contained within 'logical thinkers'. The overlapping group between 'logical thinkers' and 'analysts' could consist entirely of individuals who are not developers. Therefore, the connection is not forced by the statements.

Q26. A person walks 10 m north, 6 m east, 10 m south, and 2 m west. Find the final distance and direction from the starting point.

Solution: Net vertical displacement = 10m (North) - 10m (South) = 0 m. Net horizontal displacement = 6m (East) - 2m (West) = 4m (East). The final position is exactly **4 m East** from the starting point.

I. Data Interpretation

Q27 & Q28 Data Context: Table below outlines performance figures across training streams.

Department	Trained	Placed	Placement %
Python	120	84	70.0%
Java	100	75	75.0%
Data Analytics	80	60	75.0%
Full Stack	150	105	70.0%
Total	450	324	72.0%

Q27. Which department or departments have the highest placement percentage?

Solution: As computed in the table, both **Java** and **Data Analytics** have the highest placement rate of **75%** (75/100 and 60/80 respectively).

Q28. Find the overall placement percentage across all departments.

Solution: Total Trained = 120 + 100 + 80 + 150 = 450. Total Placed = 84 + 75 + 60 + 105 = 324. Overall Percentage = (324 / 450) × 100% = **72%**.

J. Clocks & Critical Thinking

Q29. Find the smaller angle between the clock hands at 4:40.

Solution: Using the standard angle formula: Angle = |30H - (11/2)M|, where H = 4 and M = 40. Angle = |30(4) - (11/2)(40)| = |120 - 220| = |-100| = **100°**.

Q30. Four people take 1, 2, 7, and 10 minutes to cross a bridge. They share one torch, and at most two people can cross at once. Find the minimum time required for all four to cross.

Solution: Let the people be designated by their times: 1, 2, 7, and 10.
Step 1: 1 and 2 cross over (takes 2 mins). [Total elapsed = 2 mins]
Step 2: 1 returns with the torch (takes 1 min). [Total elapsed = 3 mins]
Step 3: 7 and 10 cross over together (takes 10 mins). [Total elapsed = 13 mins]
Step 4: 2 returns with the torch (takes 2 mins). [Total elapsed = 15 mins]
Step 5: 1 and 2 cross back over together (takes 2 mins). [Total elapsed = **17 minutes**].